

YIMO Finals Solutions

August 2026

Div 2 Test A

1. Let σ denote reflection in the line ℓ . Since ℓ is the internal bisector of $\angle A$, σ fixes A and maps ray AB onto ray AC . Let $B' = \sigma(B)$, so B' lies on line AC with $AB' = AB$ and let $C' = \sigma(C)$, so C' lies on line AB with $AC' = AC$. Since $BD \perp \ell$ and $D \in \ell$, the point D is the midpoint of BB' . Similarly E is the midpoint of CC' . Let M be the midpoint of BC . In $\triangle BB'C$, the segment DM joins the midpoints of BB' and BC , so $DM \parallel B'C$. But B' and C both lie on line AC , hence $B'C = AC$ (as lines) and $DM \parallel AC$. Symmetrically, in $\triangle CC'B$, EM joins the midpoints of CC' and CB , so $EM \parallel C'B = AB$. Therefore the line through D parallel to AC and the line through E parallel to AB both pass through M . Since $AB \nparallel AC$, these two lines are distinct, so M is their unique common point.

Proposed by Shining, solution by Karam.

2. Let $n = \prod_{i=1}^k p_i^{a_i}$, where the p_i are distinct primes and $a_i \geq 1$. Then

$$d(n) = \prod_{i=1}^k (a_i + 1), \quad d(n^2) = \prod_{i=1}^k (2a_i + 1).$$

Hence the given condition is

$$\prod_{i=1}^k (2a_i + 1) = 3 \prod_{i=1}^k (a_i + 1).$$

The LHS is odd, so the RHS is odd. Thus

$$\prod_{i=1}^k (a_i + 1)$$

is odd, which implies that every a_i is even. Therefore $a_i \geq 2$, and

$$\frac{2a_i + 1}{a_i + 1} \geq \frac{5}{3}.$$

So,

$$3 = \prod_{i=1}^k \frac{2a_i + 1}{a_i + 1} \geq \left(\frac{5}{3}\right)^k.$$

Since

$$\left(\frac{5}{3}\right)^3 = \frac{125}{27} > 3,$$

we have $k \leq 2$.

If $k = 1$, then

$$\frac{2a_1 + 1}{a_1 + 1} = 3,$$

which gives $a_1 = -2$, impossible. Hence $k = 2$.

Let the two exponents be a, b . Then

$$(2a + 1)(2b + 1) = 3(a + 1)(b + 1).$$

Expanding,

$$4ab + 2a + 2b + 1 = 3ab + 3a + 3b + 3,$$

so

$$ab - a - b - 2 = 0,$$

and therefore

$$(a - 1)(b - 1) = 3.$$

Since a, b are positive integers,

$$\{a, b\} = \{2, 4\}.$$

Thus every valid n has the form

$$n = p^2 q^4,$$

where p, q are distinct primes.

Now impose $n \leq 2026$.

If $q = 2$, then

$$16p^2 \leq 2026,$$

so

$$p \in \{3, 5, 7, 11\},$$

giving

$$144, 400, 784, 1936.$$

If $q = 3$, then

$$81p^2 \leq 2026,$$

so

$$p \in \{2, 5\},$$

giving

$$324, 2025.$$

If $q \geq 5$, then

$$625p^2 > 2026,$$

so there are no further possibilities.

Hence the required sum is

$$144 + 400 + 784 + 1936 + 324 + 2025 = 5613.$$

Therefore, the last three digits are $\boxed{613}$.

Proposed by Karam, solution by Vihaan.

3. See that

$$\coth(2\theta) = \frac{\cosh(2\theta)}{\sinh(2\theta)} = \frac{\cosh^2 \theta + \sinh^2 \theta}{2 \sinh \theta \cosh \theta} = \frac{1}{2} \left(\frac{\cosh \theta}{\sinh \theta} + \frac{\sinh \theta}{\cosh \theta} \right) = \frac{1}{2} \left(\coth \theta + \frac{1}{\coth \theta} \right)$$

so $f(\coth(\theta)) = \coth(2\theta)$ thus

$$f^{(n)}(\coth(\theta)) = \coth(2^n \theta).$$

We must find θ such that $\coth(\theta) = 2$ so

$$\frac{e^\theta + e^{-\theta}}{e^\theta - e^{-\theta}} = 2 \implies e^\theta + e^{-\theta} = 2e^\theta - 2e^{-\theta} \implies 3e^{-\theta} = e^\theta \implies e^{2\theta} = 3 \implies \theta = \frac{1}{2} \ln(3).$$

Now we find

$$f^{(2026)}(2) = \coth(2^{2026} \cdot \theta) = \coth\left(2^{2026} \cdot \frac{1}{2} \ln 3\right) = \coth(2^{2025} \ln 3)$$

so since $\coth(y) = \frac{e^{2y} + 1}{e^{2y} - 1}$ we have

$$f^{(2026)}(2) = \frac{e^{2 \cdot 2^{2025} \ln 3} + 1}{e^{2 \cdot 2^{2025} \ln 3} - 1} = \frac{e^{2^{2026} \ln 3} + 1}{e^{2^{2026} \ln 3} - 1}$$

which simplifies nicely to

$$f^{(2026)}(2) = \frac{3^{2^{2026}} + 1}{3^{2^{2026}} - 1} \implies 3 + 2 + 2026 + 1 = 20\boxed{32}.$$

There is an alternate solution that avoids the use of trigonometry. Let x_n be the result of applying f to 2 n times. Thus we have the following recursion

$$x_n = \frac{x_{n-1}^2 + 1}{2x_{n-1}}.$$

See that

$$x_n - 1 = \frac{x_{n-1}^2 + 1}{2x_{n-1}} - 1 = \frac{x_{n-1}^2 - 2x_{n-1} + 1}{2x_{n-1}} = \frac{(x_{n-1} - 1)^2}{2x_{n-1}}$$

and

$$x_n + 1 = \frac{x_{n-1}^2 + 1}{2x_{n-1}} + 1 = \frac{x_{n-1}^2 + 2x_{n-1} + 1}{2x_{n-1}} = \frac{(x_{n-1} + 1)^2}{2x_{n-1}}$$

thus

$$\frac{x_n - 1}{x_n + 1} = \frac{(x_{n-1} - 1)^2}{(x_{n-1} + 1)^2} = \left(\frac{x_{n-1} - 1}{x_{n-1} + 1} \right)^2$$

let $y_n = \frac{x_n - 1}{x_n + 1}$ so $y_n = y_{n-1}^2$ thus we have

$$y_n = y_0^{2^n}.$$

See that

$$y_0 = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

thus

$$y_n = \left(\frac{1}{3} \right)^{2^n} = \frac{1}{3^{2^n}}$$

now we solve for x_n . We have the following equation

$$\frac{x_n - 1}{x_n + 1} = \frac{1}{3^{2^n}}$$

let $K = 3^{2^n}$ so

$$K(x_n - 1) = x_n + 1 \implies Kx_n - K = x_n + 1 \implies x_n(K - 1) = K + 1$$

thus

$$x_n = \frac{K + 1}{K - 1} = \frac{3^{2^n} + 1}{3^{2^n} - 1}.$$

Now we plug in the value of 2026 and arrive at the same answer as we did in the previous solution.

Proposed by Karam, solution by Karam.

4. Think of the people as vertices of a graph, with an edge for each handshake. Since everyone shakes hands with exactly k people, every vertex has degree k .

By the handshaking lemma,

$$nk = 2|E|,$$

so nk must be even. Also,

$$1 \leq k \leq n - 1.$$

Put the n people around a circle.

If $k = 2r$ is even, have each person shake hands with the r nearest people on each side. Then everyone has degree $2r = k$.

If $k = 2r + 1$ is odd, then nk even forces n to be even. Connect each person to the r nearest people on each side and also to the person directly opposite them. Then everyone has degree $2r + 1 = k$.

Thus the possible values are exactly

$$\boxed{1 \leq k \leq n - 1, \quad nk \text{ even.}}$$

So if n is even, every $k = 1, \dots, n - 1$ works. If n is odd, only even k work.
Proposed by Shining, solution by Neil.

Div 2 Test B

1. For any integer $n \geq 2$, consider writing n and 1 in base n . We have

$$n = (10)_n, \quad 1 = (1)_n,$$

so both have digit sum 1. Thus every positive integer $n \geq 2$ is friendly with 1.

Now take any $1 \leq a < b \leq 2026$. If $a = 1$, then a and b are already friendly. Otherwise,

$$a, 1, b$$

is a sequence where each consecutive pair is friendly. Hence every pair a, b is distantly friendly.

Therefore there are no pairs which are not distantly friendly, so the answer is

$$\boxed{0}.$$

Proposed by Shining, solution by Neil.

2. Let Γ be the circumcircle of $\triangle JFK$, and let Ω be the circumcircle of $\triangle XYD$. Since $JD \perp FK$ and $XY \perp JD$, we have $XY \parallel FK$. Since $\ell \parallel FK$, it follows that $XY \parallel \ell$. Let M be the midpoint of XY . Since O is the circumcenter of $\triangle JFK$, the perpendicular from O to the chord XY bisects XY . Hence $OM \perp XY$, so $OM \perp \ell$. Let S be the circumcenter of $\triangle XYD$. Since S lies on the perpendicular bisector of XY , and OM is also the perpendicular bisector of XY , we have $S \in OM$. Thus $OS \perp \ell$. Now P, Q are the intersections of Ω with ℓ , so PQ is a chord of Ω . Since S is the center of Ω , the perpendicular from S to PQ bisects PQ . But $SO \perp \ell$ and $O \in \ell$, so SO is the perpendicular bisector of PQ . Therefore O is the midpoint of PQ . Hence O is the midpoint of PQ .

Proposed by Shining, solution by Vihaan.

3. Notice the first few terms is 1, 1, 3, 11, 41, 153. Therefore, we claim $a_n = 4a_{n-1} - a_{n-2}$. Since $a_n a_{n-2} = a_{n-1}^2 + 2$, we have $a_{n+1} = \frac{a_n^2 + 2}{a_{n-1}} =$

$\frac{a_n^2 + a_n a_{n-2} - a_{n-1}^2}{a_{n-1}} = \frac{a_n(a_n + a_{n-2}) - a_{n-1}^2}{a_{n-1}} = 4a_n - a_{n-1}$. Therefore, the claim follows by induction, and all a_n are positive integers.

Proposed by Vihaan, solution written by Rayoon.

4. The arithmetic mean m of a balanced subset must also be an integer $1 \leq m \leq 2026$. We can partition the set of all balanced subsets $\mathcal{B}$ into two disjoint groups $\mathcal{B}_m$, where $\mathcal{B}_m$ consists of all non-empty subsets of S that have an arithmetic mean of m . So our sum is

$$\sum_{A \in \mathcal{B}} (-1)^{|A|} = \sum_{m=1}^{2026} \left(\sum_{A \in \mathcal{B}_m} (-1)^{|A|} \right)$$

Let's fix an integer mean $m \in \{1, 2, \dots, 2026\}$. For a subset A to have a mean of m , the sum of its elements must be $m \cdot |A|$. If $m \notin A$, we add m to the subset so $A^* = A \cup \{m\}$ and if $m \in A$, we remove m from the subset so $A^* = A \setminus \{m\}$. This operation perfectly pairs every subset $A \in \mathcal{B}_m$ with another subset A^* such that their sizes differ by exactly 1. Because their sizes differ by 1, their parities are opposite, meaning

$$(-1)^{|A|} + (-1)^{|A^*|} = 0$$

So

$$\sum_{A \in \mathcal{B}_m \cup \{\emptyset\}} (-1)^{|A|} = 0$$

Including the empty-set we get

$$1 + \sum_{A \in \mathcal{B}_m} (-1)^{|A|} = 0 \implies \sum_{A \in \mathcal{B}_m} (-1)^{|A|} = -1$$

This tells us that for any integer mean m , the sum of $(-1)^{|A|}$ over all non-empty subsets with that mean is exactly -1 . SO the sum is just

$$\sum_{A \in \mathcal{B}} (-1)^{|A|} = \sum_{m=1}^{2026} (-1) = -2026$$

Which is

$$-2026 \equiv \boxed{974} \pmod{1000}$$

Proposed by Karam, solution written by Karam.

Div 1 Test A

1. Let O_1, O_2, O_3 be the centers of $\omega_1, \omega_2, \omega_3$, respectively. Since $P, A \in \omega_1$, we have $O_1P = O_1A$. Similarly, since $P, A \in \omega_2$, we have $O_2P = O_2A$. Hence both O_1 and O_2 are equidistant from P and A , so O_1O_2 is the perpendicular bisector of PA . Thus A is the reflection of P across O_1O_2 . Similarly, B is the reflection of P across O_2O_3 , and C is the reflection of P across O_1O_3 . Now O_1, O_2, O_3

lie on γ , and P also lies on γ . Hence P lies on the circumcircle of $\triangle O_1O_2O_3$. By Simson's theorem, the perpendicular feet from P onto the three sides O_1O_2 , O_2O_3 , and O_1O_3 are collinear. Let these feet be X, Y, Z , respectively. Since A is the reflection of P across O_1O_2 , X is the midpoint of PA . Likewise, Y and Z are the midpoints of PB and PC , respectively. Therefore the homothety centered at P with ratio 2 maps X, Y, Z to A, B, C , respectively. Since a homothety maps a line not passing through its center to a parallel line, the collinearity of X, Y, Z implies the collinearity of A, B, C . Hence A, B, C are collinear.

Proposed by Karam, solution written by Vihaan.

2. Let

$$y_n = \frac{x_n}{6}.$$

Then

$$y_0 = 1, \quad y_1 = 3, \quad y_{n+1} = 6y_n - y_{n-1}.$$

The characteristic roots are $3 \pm 2\sqrt{2}$, so

$$y_n = \frac{(3 + 2\sqrt{2})^n + (3 - 2\sqrt{2})^n}{2}.$$

Since

$$(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 1,$$

we obtain

$$y_{2n} = 2y_n^2 - 1.$$

Hence

$$y_{2n} - 1 = 2(y_n - 1)(y_n + 1).$$

Let

$$a_k = v_2(y_{2^k} - 1).$$

Since

$$y_2 = 2y_1^2 - 1 = 17,$$

we have

$$a_1 = v_2(16) = 4.$$

Also, for $k \geq 1$,

$$y_{2^k} \equiv 1 \pmod{8}.$$

Indeed, this holds for $y_2 = 17$, and

$$y_{2^{k+1}} = 2y_{2^k}^2 - 1 \equiv 1 \pmod{8}$$

whenever y_{2^k} is odd. Thus

$$v_2(y_{2^k} + 1) = 1.$$

Therefore

$$a_{k+1} = v_2(2(y_{2^k} - 1)(y_{2^k} + 1)) = a_k + 2.$$

Since $a_1 = 4$,

$$a_k = 2k + 2.$$

Thus

$$v_2(x_{2^{2026}} - 6) = v_2(6(y_{2^{2026}} - 1)) = 1 + a_{2026} = 1 + (2 \cdot 2026 + 2) = 4055.$$

Therefore the last three digits of x are

$$\boxed{055}.$$

Proposed by Karam, Solution written by Neil

3.

Let the sum be S so

$$S = \frac{a^2}{\sqrt{b+c}} + \frac{b^2}{\sqrt{c+a}} + \frac{c^2}{\sqrt{a+b}}$$

We apply Hölder's Inequality to $\left(\frac{a^2}{\sqrt{b+c}}, \frac{b^2}{\sqrt{c+a}}, \frac{c^2}{\sqrt{a+b}}\right)$, $\left(\frac{a^2}{\sqrt{b+c}}, \frac{b^2}{\sqrt{c+a}}, \frac{c^2}{\sqrt{a+b}}\right)$, and $(a^2(b+c), b^2(c+a), c^2(a+b))$ to get

$$\left(\sum_{cyc} \frac{a^2}{\sqrt{b+c}}\right) \left(\sum_{cyc} \frac{a^2}{\sqrt{b+c}}\right) \left(\sum_{cyc} a^2(b+c)\right) \geq \left(\sum_{cyc} a^2\right)^3$$

this simplifies to

$$S^2 \sum_{cyc} a^2(b+c) \geq (a^2 + b^2 + c^2)^3$$

To show that $S \geq \frac{3}{\sqrt{2}}$, it suffices to show that $S^2 \geq \frac{9}{2}$. Substituting our bound for S^2 , it is sufficient to prove that

$$\frac{(a^2 + b^2 + c^2)^3}{\sum_{cyc} a^2(b+c)} \geq \frac{9}{2} \implies 2(a^2 + b^2 + c^2)^3 \geq 9 \sum_{cyc} a^2(b+c)$$

See that $2(a^2 + b^2 + c^2)^3 \geq 9 \sum_{cyc} a^2(b+c)$ is not homogeneous; the left side has degree 6, while the right side has degree 3 so we use $a^3 + b^3 + c^3 + abc = 4$ to homogenize it. Because $\frac{a^3 + b^3 + c^3 + abc}{4} = 1$, we can multiply the RHS by this expression without changing its value

$$2(a^2 + b^2 + c^2)^3 \geq 9 \left(\sum_{cyc} a^2(b+c)\right) \left(\frac{a^3 + b^3 + c^3 + abc}{4}\right)$$

Multiplying both sides by 4 gives

$$8(a^2 + b^2 + c^2)^3 \geq 9(a^3 + b^3 + c^3 + abc) \sum_{cyc} (a^2b + a^2c)$$

We use the symmetric sum notation $T(u, v, w) = \sum_{sym} a^u b^v c^w$, which represents the sum over all 6 permutations of (a, b, c) for the rest of the problem. First expand the LHS

$$(a^2 + b^2 + c^2)^3 = a^6 + b^6 + c^6 + 3 \sum_{sym} a^4 b^2 + 6a^2 b^2 c^2$$

which is

$$(a^2 + b^2 + c^2)^3 = \frac{1}{2}T(6, 0, 0) + 3T(4, 2, 0) + T(2, 2, 2)$$

$$\text{LHS} = 8 \left(\frac{1}{2}T(6, 0, 0) + 3T(4, 2, 0) + T(2, 2, 2) \right) = 4T(6, 0, 0) + 24T(4, 2, 0) + 8T(2, 2, 2)$$

Now the RHS is

$$\sum_{cyc} (a^2 b + a^2 c) = \sum_{sym} a^2 b = T(2, 1, 0)$$

$$a^3 + b^3 + c^3 = \frac{1}{2}T(3, 0, 0)$$

$$abc = \frac{1}{6}T(1, 1, 1)$$

Therefore, the RHS is $9 \cdot T(2, 1, 0) \left(\frac{1}{2}T(3, 0, 0) + \frac{1}{6}T(1, 1, 1) \right)$ distributing this yields

$$T(2, 1, 0) \cdot \frac{1}{2}T(3, 0, 0) = T(5, 1, 0) + T(4, 2, 0) + T(3, 2, 1)$$

and

$$T(2, 1, 0) \cdot \frac{1}{6}T(1, 1, 1) = T(3, 2, 1)$$

Combining these terms gives

$$\text{RHS} = 9(T(5, 1, 0) + T(4, 2, 0) + 2T(3, 2, 1)) = 9T(5, 1, 0) + 9T(4, 2, 0) + 18T(3, 2, 1)$$

We want to prove $\text{LHS} \geq \text{RHS}$ so subtracting RHS gives

$$4T(6, 0, 0) - 9T(5, 1, 0) + 15T(4, 2, 0) + 8T(2, 2, 2) - 18T(3, 2, 1) \geq 0$$

Consider the following polynomial and its factorization:

$$4a^6 - 9a^5b + 15a^4b^2 - 20a^3b^3 + 15a^2b^4 - 9ab^5 + 4b^6 = (a-b)^2(4a^4 - a^3b + 9a^2b^2 - ab^3 + 4b^4)$$

We can verify that the second factor is strictly positive for all real (a, b) by completing the square

$$4a^4 - a^3b + 9a^2b^2 - ab^3 + 4b^4 = 4 \left(a^2 - \frac{1}{8}ab \right)^2 + b^2 \left(\frac{143}{16}a^2 - ab + 4b^2 \right)$$

The discriminant of the quadratic inside the second term is $1 - 4\left(\frac{143}{16}\right)(4) = -142 < 0$ so the entire expression is non-negative. Thus, for any reals a and b

$$4(a^6 + b^6) - 9(a^5b + ab^5) + 15(a^4b^2 + a^2b^4) \geq 20a^3b^3$$

Summing this inequality cyclically over (a, b) , (b, c) , (c, a) yields exactly the symmetric sums

$$4T(6, 0, 0) - 9T(5, 1, 0) + 15T(4, 2, 0) \geq 20 \sum_{cyc} a^3 b^3$$

Since $20 \sum_{cyc} a^3 b^3 = 10 \sum_{sym} a^3 b^3 = 10T(3, 3, 0)$, we have

$$4T(6, 0, 0) - 9T(5, 1, 0) + 15T(4, 2, 0) \geq 10T(3, 3, 0)$$

Substitute the bound we previously derived into our main target inequality to get

$$[4T(6, 0, 0) - 9T(5, 1, 0) + 15T(4, 2, 0)] + 8T(2, 2, 2) - 18T(3, 2, 1) \geq 10T(3, 3, 0) + 8T(2, 2, 2) - 18T(3, 2, 1)$$

Now we need to prove that $10T(3, 3, 0) + 8T(2, 2, 2) \geq 18T(3, 2, 1)$. By Schur's Inequality applied to the variables $x = ab$, $y = bc$, and $z = ca$, we have

$$x^3 + y^3 + z^3 + 3xyz \geq \sum_{sym} x^2 y$$

Mapping this back to a, b, c yields

$$x^3 + y^3 + z^3 = a^3 b^3 + b^3 c^3 + c^3 a^3 = \frac{1}{2}T(3, 3, 0)$$

$$3xyz = 3a^2 b^2 c^2 = \frac{1}{2}T(2, 2, 2)$$

$$\sum_{sym} x^2 y = \sum_{sym} (ab)^2 (bc) = \sum_{sym} a^2 b^3 c = T(3, 2, 1)$$

Substituting these gives

$$\frac{1}{2}T(3, 3, 0) + \frac{1}{2}T(2, 2, 2) \geq T(3, 2, 1)$$

Multiplying by 16 gives

$$8T(3, 3, 0) + 8T(2, 2, 2) \geq 16T(3, 2, 1)$$

We are left with an excess of $2T(3, 3, 0)$, by Muirhead's Inequality, because the sequence $(3, 3, 0)$ majorizes $(3, 2, 1)$, we have:

$$2T(3, 3, 0) \geq 2T(3, 2, 1)$$

Adding these two bounds together yields

$$8T(3, 3, 0) + 8T(2, 2, 2) + 2T(3, 3, 0) \geq 16T(3, 2, 1) + 2T(3, 2, 1)$$

$$10T(3, 3, 0) + 8T(2, 2, 2) \geq 18T(3, 2, 1)$$

Thus

$$4T(6, 0, 0) - 9T(5, 1, 0) + 15T(4, 2, 0) + 8T(2, 2, 2) \geq 10T(3, 3, 0) + 8T(2, 2, 2) \geq 18T(3, 2, 1)$$

Which confirms our target inequality

$$4T(6, 0, 0) - 9T(5, 1, 0) + 15T(4, 2, 0) + 8T(2, 2, 2) - 18T(3, 2, 1) \geq 0$$

and we are done. Equality holds when $a = b = c = 1$.

Proposed by Karam, solution by Karam.

4. The idea is to rephrase the problem in the following way. Suppose a counter is at 0. An unfair coin is flipped: if it comes tails the counter goes +1, and if it comes heads the counter goes -2. Calculate the expected amount of times that the counter hits 0. Now this invites us to define a function $f(n)$ which is the expected amount of times that the counter hits 0 starting from a number n and this allows us to use a states type argument. Now for most values of n we have that $f(n) = (f(n+1) + 2f(n-2))/3$ the exception is $n = -1$ and $n = 2$, where we have $f(-1) = (f(0) + 1 + 2f(-3))/3$ and $f(2) = (f(3) + 2f(0) + 2)/3$ because hitting 0 adds 1. Now consider f of negative values because of the $f(n) = (f(n+1) + 2f(n-2))/3$ recursion (which has characteristic polynomial $x^3 - 3x^2 + 2$) we know that there exists A, B, C such that for all negative integers n , $f(n) = A + B(1 + \sqrt{3})^n + C(1 - \sqrt{3})^n$. We know that f should tend to 0 as n goes to $-\infty$, this forces $A = 0$ and $C = 0$. So $f(n) = B(1 + \sqrt{3})^n$ for positive integers p by the same reasoning we must also have $f(p) = A + B(1 + \sqrt{3})^p + C(1 - \sqrt{3})^p$ for some constants A, B, C . We know that $f(p)$ should tend to some constant value as p goes to infinity so this forces $B = 0$ so $f(p) = A + C(1 - \sqrt{3})^p$. Now all that's left is solving for A, B, C , and whatever $f(0)$ is. Using $f(-1) = (f(0) + 1 + 2f(-3))/3$ we get that $f(0) = B - 1$, using $f(2) = (f(3) + 2f(0) + 2)/3$ we get that $f(0) = A + C - 1$. This means $B = A + C$ and we just solve for A and C using two more equations $3f(0) = f(1) + 2f(-2)$ and $3f(1) = f(2) + 2f(-1)$. You get that $A = 1$, $C = (\sqrt{3} - 1)/2$, consequently the answer is $A + C - 1 = (\sqrt{3} - 1)/2$. Therefore $a = 3$, $b = 1$, and $c = 2$, so $a + b + c = \boxed{6}$.

Proposed by Rayoon, solution written by Rayoon.

Div 1 Test B

1. The trick is to recognize the expression as a quadratic form with a very clean upper bound.

$$\text{Set } a = x, \quad b = 2y, \quad c = 3z.$$

$$\text{Then } a^2 + b^2 + c^2 = k \text{ and } (x+y)(y+3z) = \left(a + \frac{b}{2}\right) \left(\frac{b}{2} + c\right).$$

$$\text{Expanding, } (x+y)(y+3z) = \frac{1}{2}ab + ac + \frac{1}{4}b^2 + \frac{1}{2}bc.$$

$$\text{Now observe the identity } \frac{3}{4}(a^2 + b^2 + c^2) - \left(\frac{1}{2}ab + ac + \frac{1}{4}b^2 + \frac{1}{2}bc\right) = \frac{1}{4}(a - b)^2 + \frac{1}{2}(a - c)^2 + \frac{1}{4}(b - c)^2.$$

$$\text{Hence } (x+y)(y+3z) \leq \frac{3}{4}(a^2 + b^2 + c^2) = \frac{3k}{4}.$$

Equality occurs when $a = b = c$, i.e. $x = 2y = 3z$, which is allowed since $x, y, z > 0$.

Therefore the maximum possible value is exactly $\frac{3k}{4}$.

We are told this maximum is an integer, so $\frac{3k}{4} \in \mathbb{Z}$.

Since $\gcd(3, 4) = 1$, this happens exactly when $4 \mid k$.

The positive multiples of 4 from 1 through 10^3 are $\frac{10^3}{4} = \boxed{250}$.

Proposed by Shining, solution by Vihaan.

2. Work modulo 103 throughout.

Let $P(x) = \prod_{k=1}^{102} (k^2 + x)$. Write $P(x) = x^{102} + c_{101}x^{101} + \dots + c_0$.

For $0 \leq m < 102$, $\sum_{a=1}^{102} a^m \equiv -1 \pmod{103}$, while $\sum_{a=1}^{102} a^{102} \equiv 1 \pmod{103}$.

Therefore $\sum_{a=1}^{102} P(a) \equiv 1 - (c_{101} + \dots + c_0)$.

Since $P(1) = 1 + c_{101} + \dots + c_0$, we obtain $\sum_{a=1}^{102} P(a) \equiv 2 - P(1) \pmod{103}$.

It remains to find $P(1)$.

Let R be the set of nonzero quadratic residues modulo 103. Since every nonzero quadratic residue occurs twice among $1^2, 2^2, \dots, 102^2$, we have $P(1) = \left(\prod_{r \in R} (r + 1)\right)^2$.

Now, by Fermat's little theorem, every $r \in R$ satisfies $r^{51} = 1 \pmod{103}$.

There are exactly 51 such residues, so $x^{51} - 1 = \prod_{r \in R} (x - r) \pmod{103}$.

Setting $x = -1$ gives $(-1)^{51} - 1 = \prod_{r \in R} (-1 - r)$, so $-2 = -\prod_{r \in R} (1 + r)$. Hence $\prod_{r \in R} (1 + r) \equiv 2 \pmod{103}$.

Therefore $P(1) \equiv 2^2 \equiv 4 \pmod{103}$.

Thus $\sum_{a=1}^{102} P(a) \equiv 2 - 4 \equiv -2 \equiv 101 \pmod{103}$.

Hence the required remainder is $\boxed{101}$.

Proposed by Karam, Solution written by Vihaan.

3.

$$B = (-1, 0), \quad C = (1, 0), \quad A = (u, v).$$

Then

$$H = \left(u, \frac{1 - u^2}{v}\right), \quad P = \left(u, \frac{u^2 - 1}{v}\right).$$

Since the perpendicular bisector of BC is the y -axis,

$$O = (0, s), \quad s = \frac{u^2 + v^2 - 1}{2v}.$$

Now EF is the line $y = v/2$, so

$$Q = (0, v - s) = \left(0, \frac{v^2 - u^2 + 1}{2v}\right).$$

Also

$$G = \left(\frac{u}{3}, \frac{v}{3}\right).$$

Therefore

$$P + 2Q = \left(u, \frac{u^2 - 1}{v} + \frac{v^2 - u^2 + 1}{v}\right) = (u, v) = 3G.$$

Hence P, G, Q are collinear.

Let N be the center of the nine-point circle. Since N is the midpoint of O and H ,

$$N = \frac{O + H}{2}.$$

Writing $Q = (0, q)$, we have

$$q = v - s = \frac{v^2 - u^2 + 1}{2v} = s + \frac{1 - u^2}{v},$$

so

$$N = \left(\frac{u}{2}, \frac{q}{2}\right).$$

Thus

$$QN^2 = \frac{u^2 + q^2}{4}.$$

But

$$u^2 + q^2 = u^2 + (v - s)^2 = 1 + s^2 = R^2,$$

where R is the circumradius. Hence

$$QN = \frac{R}{2},$$

which is the radius of the nine-point circle. Therefore Q lies on the nine-point circle.

Proposed by Abhi, solution written by Abhi.

4. Make a bipartite graph: on the left, two copies of each committee; on the right, the people. Join a copy of a committee to its members.

Take k committees, and let m be the number of people in their union. For each person x , let r_x be the number of these committees containing x . Then

$$\sum_x r_x = k(2n + 1).$$

Since two committees have at most one common member,

$$\sum_x \binom{r_x}{2} \leq \binom{k}{2}.$$

Hence

$$\sum_x r_x^2 = \sum_x r_x + 2 \sum_x \binom{r_x}{2} \leq k(2n + 1) + k(k - 1) = k(2n + k).$$

By Cauchy–Schwarz,

$$k^2(2n+1)^2 = \left(\sum_x r_x \right)^2 \leq m \sum_x r_x^2 \leq mk(2n+k).$$

Thus

$$m \geq \frac{k(2n+1)^2}{2n+k}.$$

Since $k \leq n^2$,

$$(2n+1)^2 > 2(2n+k),$$

so

$$m > 2k.$$

Thus any k committees contain at least $2k$ people. Any set of officer slots coming from k committees has at most $2k$ slots, so Hall's condition holds. Therefore there is a matching covering all $2n^2$ slots, giving a distinct chair and vice-chair for every committee.

Proposed by Abhi, solution written by Abhi.